

Teleoperated Robot Sells Toothbrush in a Shopping Mall: A Field Study

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ABSTRACT

Social robots have been used to provide services in many public spaces. However, many of them are autonomous robots that have poor natural language conversation skills. It is considered a difficult task for an autonomous robot to be a salesperson. In this paper, we propose the use of a teleoperated robot with a human operator for sales purposes. We conducted an exploratory field experiment in which we built a sales booth in a shopping mall and used our teleoperated robot as the salesperson to promote the sales of toothbrushes. We found that our robot has the ability to attract people's attention and influence their shopping behavior. Furthermore, we discuss the research challenges in designing and applying teleoperated robots to real sales situations.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

teleoperated robot, field study, sales, conversation, human-robot interaction

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1 INTRODUCTION

Robots play a significant role in our daily lives. Other than industrial robots that work in factories, social robots are increasingly being deployed in both public spaces such as train stations [3], shopping

malls [5, 15], and hotels [7, 9], and domestic environments such as people's homes [6]. A common feature of such social robots is that they are able to communicate with humans in natural ways. A social robot can talk in human language, use gaze, and perform gestures. Such natural communication ability enables robots to work on tasks such as guiding [4], providing recommendations [12], or simply chatting for fun [1].

The majority of social robots are autonomous, meaning that their behaviors are managed by computer programs and their conversations with people rely on speech recognition and synthesis. However, social robots are often deployed in public spaces, and such noisy environments inhibit speech recognition [16]. In addition, current speech technology has still not achieved high-levels of natural conversation. Therefore, to overcome such limitations, some of these robots are partially operated by a human staff whose role is to help the robots decide what to say during a conversation [12].

Teleoperated robots, by contrast, rely on human operators. They are often used for telepresence [14] or remote work [13]. For instance, they allow people to work remotely from home and perform tasks in different locations. In particular, because a human operator manages speech processing instead of a computer program, it enables teleoperated robots to talk naturally with people. Thus, teleoperated robots are a better choice for tasks that require high-level natural language conversation. For example, [1] reported a case study in which they developed a teleoperated robot system for a festival, and their robots were used for welcoming and product recommendation.

Selling is a task that requires high-level natural language communication. For traditional counter selling in a store, a salesperson would expect to build social relationships with his or her customers through conversation [10]. Hence, selling is regarded as a difficult task for robots. Previous studies have investigated using social robots to hand out fliers [11], recommend shops in shopping malls [12], and provide guiding services [2], but few explored how social robots could sell products. Watanabe et al. [16] put an android in a department store and found that their robot sold 43 sweaters within 10 days. Although their android looked like a real person, the conversation system was mainly touch-display based because a precise voice recognition system was needed.

We consider that there are two advantages to applying teleoperated robots in sales positions. First, teleoperated robots are able to

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Figure 1: Overview of our sales booth

achieve high-level conversations with their customers. This allows the robots to build social relationships with customers. Second, tele-operated robots are capable of dealing with various situations that happened during the task and applying proper selling strategies accordingly.

In this paper, we explore the question of how a teleoperated robot sells goods in the real world. We developed a system that allows a person to operate and talk through a robot without requiring any special technical skills. We put the robot in a shopping mall and let it sell one kind of toothbrush (KISEKINO toothbrush, approximately 6 dollars per toothbrush) for four hours. As a result, our robot successfully sold eight toothbrushes, which is significantly more than the average number of 1 per day according to the data obtained from a drug store (a typical Japanese drug store sells many kinds of toothbrushes). In addition, we found that many of the customers approached the robot first and interacted with it, which shows the potential merit of using a robot for the selling task. Our observations provide important knowledge for designing teleoperated robots as salespersons and to serve new roles in the future.

2 FIELD STUDY

2.1 Task

We managed to build our own sales booth in the common area of a shopping mall in Osaka, Japan. For the commodity, we considered a toothbrush; we were able to promote and sell a new toothbrush product named KISEKINO. This toothbrush is about five times more

expensive than normal ones and, according to the data we received from a drug store, the average sales is approximately 1 toothbrush per day for that store. Therefore, a key factor in the sale of such a product is communicating with customers and explaining its unique features compared to other toothbrushes.

The experiment lasted from 12 : 00 to 17 : 00 on a Sunday; a 1 hour break was set between 14 : 00 and 15 : 00. We employed a university student (female, native Japanese) as the operator of our robot. She is extroverted and active. We believe that she was well suited to this role because selling requires active conversation with customers. An experimenter was hidden and sitting behind the booth during the experiment and only interacted with the customers who decided to buy the toothbrush to complete the payment.

In this study, the task of the teleoperated robot was to greet the passerby, promote the KISEKINO toothbrush, and persuade them to buy the product. To build social relationships with customers, a subtask was to converse with them.

2.2 System

Figure 1 shows the overview of our sales booth. We put some sample toothbrushes in the middle of the booth to allow customers to easily pick one up and evaluate the features of the product. We installed a 13 inch display to the left side of the booth through which the operator of our robot was able to play promotion videos and present slides for information.

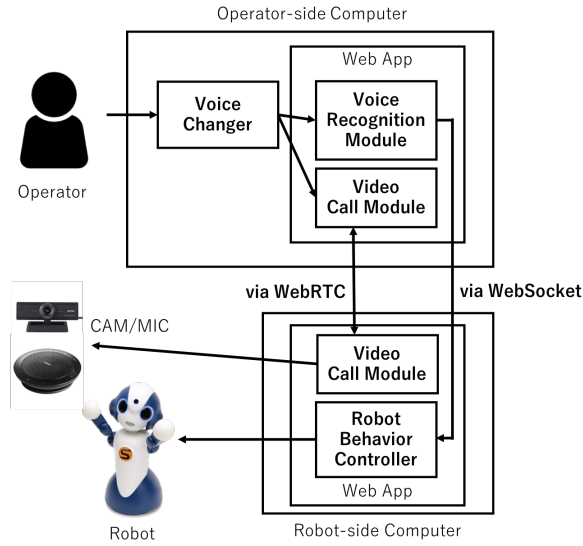


Figure 2: System architecture of our teleoperated system

A small humanoid robot, called “SOTA” (Vstone Co., Ltd.), is located on the left-hand side of the booth. The operator can remotely control the robot via a video call web application and talk through a speakerphone (in front of the box located under the robot). The operator watches the video from the ultra-wide-angle web camera located at the back of the robot). A mini PC set inside the box ran the program for the video call web application and sent control commands to the robot. Figure 2 shows the architecture of our developed teleoperated system. In addition, we had a battery and a WiFi router inside the wagon that held the robot. The battery provided approximately 6 h of power supply for the robot, and the WiFi router allowed the robot to connect to the Internet.

2.3 Strategy

We told the operator to follow a basic operation flow:

- Step 1 : Greet passerby and promote the KISEKINO toothbrush. An example could be “Hello! I am here selling the KISEKINO toothbrush. It is a very good product!”
- Step 2 : When someone stops, persuade him or her to pick up a sample toothbrush and take a close look at it. Go to step 3 if the customer shows no reaction of picking up the toothbrush.
- Step 3 : Play the promotion video (30 seconds) and explain the advantages of the toothbrush.
- Step 4 : Say something to urge the customer to buy the toothbrush. An example could be “I really hope you could buy it!”
- Step 5 : If the customer decides to buy the toothbrush, tell him or her to ask the staff behind the booth for payment.

In addition to this operation flow, the operator could freely choose what to say and conduct small talk with the customer according to the situation.

2.4 Evaluation

The entire experiment was video recorded for analysis. To explore the effects of our teleoperated robot on the selling task, we coded the video data and counted the following items:

- Total number of passersby
- Number of people who stopped (stayed at the sales booth for more than 1 second)
- Time duration of the people who stayed at the sales booth (except for those who bought the toothbrush because payment process costed several minutes)
- Number of people who stopped and were in a group
- Number of people who stopped in front of the robot first (instead of in front of the toothbrush)
- Number of people who picked up a sample toothbrush
- Number of people who picked up a sample toothbrush when the robot said “Please pick up a toothbrush and take a look at it.”
- Number of people who bought the toothbrush

Note that the numbers were counted in number of groups instead of persons, meaning that, for example, a couple of two persons who stopped at the sales booth was counted as “1”, as well as one person that stopped alone.

In addition to the coding, we checked the videos and looked for interesting interactions and patterns that could reflect important points and research challenges. In addition, we had a quick discussion with the operator to gather her impressions. Basically, we addressed specific questions that “How successful can the teleoperated robot be in getting and holding on the attention of the passerby?” “What behaviors exhibited by a potential customer (e.g., picking up the product) could indicate that the person might buy the product?” and “What are the experience and impressions of the operator of the teleoperated robot?”

3 RESULTS

3.1 Coding Results

Table 1 demonstrates the results of coding. We then further analyze the results to discover the potential effects of the teleoperated robot on the customers behavior (Figure 3).

3.1.1 Stop Rate. The stop rate (SR) is the ratio of the number of people who stopped at the sales booth to the total number of passersby. SR can be calculated as $\frac{b}{a}$ ($= 11.23\%$). This indicates that approximately 1 out of 10 of the passersby stopped and stayed at the sales booth for at least 1 s. Together with coding item c (Table 1), it could be concluded that about 1 out of 10 of the passersby stopped and checked our sales booth for an average of 45 seconds. In particular, $\frac{d}{b}$ ($= 80.49\%$) reveals that the majority of people who stopped at the sales booth were in a group, such as couples or families with children, and not alone. A similar observation was described in a previous work [15].

Additionally, we want to explore the relationship between SR and the robot (SR-R). $\frac{c}{b}$ ($= 74.39\%$), indicates that, among the people who stopped at the sales booth, over 70% of them stopped in front of the robot first instead of the product.

Our teleoperated system was built on top of the WebRTC technology.

Table 1: Video data coding results

ID	Coding Item	Results
a	Total number of passersby	730
b	Number of people who stopped	82
c	Average time duration for the people who stayed at the sales booth	45s
d	Number of people who stopped and were in a group	66
e	Number of people who stopped in front of the robot first	61
f	Number of people who picked up a sample toothbrush	17
g	Number of people who picked up a sample toothbrush when asked by the robot	13
h	Number of people who bought the toothbrush	4*
h-1	Number of people who bought the toothbrush that picked up a sample toothbrush for a closer look	4

*In total 8 toothbrushes were sold

3.1.2 Pick-Up Rate. The pick-up rate (PUR) is the ratio of the number of people who picked up a sample toothbrush and had a closer look at it to the number of people who stopped at the sales booth. PUR is calculated as $\frac{f}{b}$ ($= 20.73\%$). It shows that about 1 out of 5 of the people who stopped at the sales booth picked up a sample toothbrush and checked it.

In addition, we investigated the relationship between PUR and the robot (PUR-R). PUR-R, $\frac{g}{f}$ ($= 76.47\%$), reveals that the majority of the people who picked up a sample toothbrush because the robot asked them to do so.

3.1.3 Purchase Rate. The purchase rate (PR) is the ratio of the number of people who eventually purchased the toothbrush to the number of people who stopped at the sales booth. PR is calculated as $\frac{h}{b}$ ($= 4.88\%$), indicating that approximately 5% of the people who stopped at the sales booth bought one or more toothbrushes.

It is difficult to observe the relationship between PR and the robot directly. However, it is possible to examine the effect of the sales strategy, which involves asking customers to pick up a sample toothbrush and examine it, on their purchase behavior. Hence, we further explore the relationship between PR and the pick-up behavior (PR-PU). PR-PU, $\frac{h-1}{f}$ ($= 23.53\%$), suggests that almost one out of four of the people who picked up a sample toothbrush eventually bought the toothbrush, because $\frac{h-1}{h}$ ($= 100\%$) shows that all the people who bought the toothbrush picked up a sample toothbrush and checked it before their purchase.

3.2 Operator Interview

Overall, the operator enjoyed the task of talking through the robot and selling a toothbrush. She felt physically relaxed because she was sitting throughout the entire study and operating the robot did not require a significant amount of work. However, the operator reported that “I felt stressed when someone passed through and ignored my words.” She felt mental stress, particularly when she tried to talk to a person but was ignored by them and got no response.

The operator found that it was important to keep calling out to the passerby to get their attention. She commented that “I tried to call out passersby using voice acting skills but talked in normal ways when the person came close to me (the robot).” Sometimes she tried to change her pitch of voice and the way of speaking to get

better effect. However, when people stopped and showed interest, she often returned to her normal way of speaking.

With regard to task performance, however, the operator said that she did not feel significantly accomplished even though she managed to sell eight toothbrushes to four groups of people, which is much higher than the average number of toothbrushes sold per day in a drug store. She reported that “I felt that people came and bought a toothbrush because of their interest in the robot but not my selling skills.”

4 DISCUSSION

4.1 Robot Increases Exposure of Product

The SR results indicate that over 10% of passerby stopped at our sales booth. Previous studies reported similar results [8]. Moreover, the SR-R shows that over 70% of the people who stopped at the sales booth came to the robot before observing the toothbrush. This confirms that the robot was able to attract the attention of passerby. Indeed, a robot in a public space would probably gather a significant amount of attention due to its novelty. This effect can be beneficial to companies by increasing the exposure of their products. Therefore, we believe that a robot could be considered an effective method for product advertising.

4.2 Importance of Pick-Up behavior

The result of PR-PU confirms the importance of pick-up behavior, as about 1 out of 4 of the people who picked up a sample eventually purchased the toothbrush. However, it should be noted that our results cannot explain the causal relationship between pick-up behavior and purchase. It is not clear whether the customers bought the toothbrush owing to the pick-up behavior or they picked up a sample toothbrush because they had a certain level of willingness to purchase at the beginning. Nevertheless, we consider pick-up behavior as a potentially important factor for sales promotion and interaction design.

4.3 Robot and Sales

In this study, our robot tried to ask customers to pick up a sample toothbrush and take a close look at its new features. The results confirmed the importance of pick-up behavior. Interestingly, the PUR-R indicates that most people (76.47%) picked up a sample toothbrush

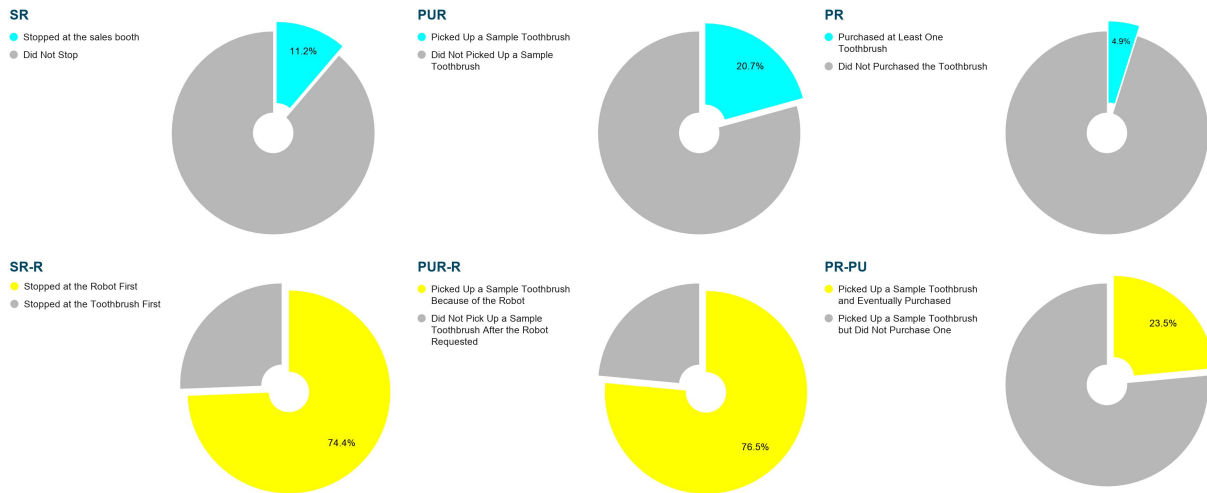


Figure 3: Summary of results

because the robot asked them to do so. This suggests the persuasive power of the robot, which could potentially influence people's behaviors. Therefore, our results indicate that it is promising to use a robot for sales promotion and provide insights for effective human robot interaction design in a real sales situation.

4.4 Autonomous vs. Teleoperated Robot

We used a teleoperated robot as a salesperson in our experiment. In contrast to the autonomous robots that were used in past studies [5, 16], our teleoperated robot was capable of natural conversation with people because an operator was talking through the robot. Hence, our robot successfully managed to deal with various situations and adjust its speaking timing and contents to the situations accordingly. For example, it was easy for the robot to decide an appropriate timing to ask the customers to pick up a sample toothbrush. Moreover, the robot was capable of chatting with people.

Therefore, although an autonomous robot could be effective in attracting passerby's attention as well, such a robot may encounter many difficulties while interacting with customers. Previous studies found that it could be difficult for a robot to outperform human staff [15]. However, because building social relationships with customers is important for sales work and conversation is critical to this relationship [16], we suggest that a teleoperated robot has the potential to achieve better performance than an autonomous robot. In other words, a teleoperated robot could achieve a performance similar to that of a human staff or even outperform the person.

4.5 Research Challenges

Based on our observations, we summarize the following research challenges in employing a teleoperated robot as a salesperson in a real sales situation:

- **Interaction Design for Sales Promotion:** In our study, one group of people came to the robot and chatted with it. However, when the robot started to promote the toothbrush, the people suddenly lost interest in the robot and left. Hence,

interactions must be designed to accommodate various contexts in order to keep people's interest.

- **Social Pressure:** A robot may exert less social pressure on potential customers compared to human staff. Therefore, it is easier for passersby to stop at a robot but leave during conversation as well. We consider relieving social pressure through the use of a robot an important research topic.
- **Operator's Stress:** According to our interview results, an operator may feel mental stress, particularly when the robot received no response from people. Such stress could influence the operator's motivation and job performance. Therefore, we think that there should be a focus on improving the operating system to reduce the operator's mental stress and maintain his or her motivation at a high level.

5 CONCLUSION

In this study, we stationed a teleoperated robot in a shopping mall and built a booth through which it sold a brand of toothbrush. We recorded videos to determine what is necessary for our robot to become a successful salesperson. We discovered that our robot has the power to attract people's attention and persuade them to pick up a sample toothbrush, which can further influence people's purchase decisions. Finally, we summarize the research challenges in designing and applying teleoperated robots in real sales situations.

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